

Creation Date 23-Jan-2009

Revision Date 22-Sep-2023

Revision Number 8

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	3-Chloroperoxybenzoic acid, 70-75%
Cat No. :	109790000
Synonyms	m-Chloroperbenzoic acid; m-CPBA; MCPBA
CAS No	937-14-4
Molecular Formula	C7 H5 Cl O3
REACH registration number	01-2120794224-51

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory/Industrial use chemical/reagent. Chemical intermediate. See Annex for full list.
Sector of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites SU9 - Manufacture of fine chemicals SU24 - Scientific research and development
Product category	PC21 - Laboratory chemicals
Process categories	PROC1 - Use in closed process, no likelihood of exposure PROC9 - Transfer of substance or mixture into small containers (dedicated filling line, including weighing) PROC15 - Use as a laboratory reagent
Environmental release category	ERC6b - Industrial use of reactive processing aids ERC8b - Wide dispersive indoor use of reactive substances in open systems ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against	Food, drug, pesticide or biocidal product use

1.3. Details of the supplier of the safety data sheet

Company

EU entity/business name
Thermo Fisher Scientific
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road,
Loughborough, Leicestershire LE11 5RG, United Kingdom

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
e-mail - infoch@thermofisher.com

E-mail address

begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**

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Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Organic peroxides

Type D (H242)

Health hazards

Acute oral toxicity
Skin Corrosion/Irritation
Serious Eye Damage/Eye Irritation
Skin Sensitization

Category 4 (H302)
Category 1 C (H314)
Category 1 (H318)
Category 1 (H317)

Environmental hazards

Acute aquatic toxicity
Chronic aquatic toxicity

Category 1 (H400)
Category 1 (H410)

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H242 - Heating may cause a fire
H302 - Harmful if swallowed
H314 - Causes severe skin burns and eye damage
H317 - May cause an allergic skin reaction
H410 - Very toxic to aquatic life with long lasting effects

Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P220 - Keep away from clothing and other combustible materials
P280 - Wear protective gloves/protective clothing/eye protection/face protection

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P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician

2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
3-Chloroperoxybenzoic acid	937-14-4	EEC No. 213-322-3	70-76	Org. Perox D (H242) Acute Tox. 4 (H302) Skin Corr. 1C (H314) Skin Sens. 1 (H317) Eye Dam. 1 (H318) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)
Benzoic acid, 3-chloro-	535-80-8	EEC No. 208-618-4	<20	Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) STOT SE 3 (H335)
Water	7732-18-5	231-791-2	>17	-

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
3-Chloroperoxybenzoic acid	-	1	-

REACH registration number	01-2120794224-51
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Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.
Skin Contact	Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.
Ingestion	Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled

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the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. May cause allergic skin reaction. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

CO₂, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

5.2. Special hazards arising from the substance or mixture

The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Hydrogen chloride gas.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

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6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Keep refrigerated. Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials. Corrosives area.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Storage Class/LGK 5.2

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local	Acute effects	Chronic effects local	Chronic effects
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	(Dermal)	systemic (Dermal)	(Dermal)	systemic (Dermal)
3-Chloroperoxybenzoic acid 937-14-4 (70-76)				DNEL = 0.75mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
3-Chloroperoxybenzoic acid 937-14-4 (70-76)				DNEL = 2.64mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
3-Chloroperoxybenzoic acid 937-14-4 (70-76)	PNEC = 0.45µg/L	PNEC = 22.2µg/kg sediment dw	PNEC = 4.5µg/L	PNEC = 2mg/L	PNEC = 4.17µg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
3-Chloroperoxybenzoic acid 937-14-4 (70-76)	PNEC = 45ng/L	PNEC = 2.22µg/kg sediment dw	PNEC = 0.45µg/L		

8.2. Exposure controls

Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber	See manufacturers	-	EN 374	(minimum requirement)
Neoprene	recommendations			
Natural rubber				
PVC				

Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Particulates filter conforming to EN 143

Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure

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limits are exceeded or if irritation or other symptoms are experienced.

Recommended half mask:- Particle filtering: EN149:2001
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	paste Solid	
Appearance	Off-white	
Odor	Slight	
Odor Threshold	No data available	
Melting Point/Range	92 - 94 °C / 197.6 - 201.2 °F	
Softening Point	No data available	
Boiling Point/Range	264 °C / 507.2 °F	Estimated
Flammability (liquid)	Not applicable	Solid
Flammability (solid,gas)	No information available	
Explosion Limits	No data available	
Flash Point	Not applicable	Method - No information available
Autoignition Temperature	No data available	
Decomposition Temperature	> 88°C	
Self-Accelerating Decomposition Temperature (SADT)	55°C	
pH	4.5	sat.aq.sol
Viscosity	Not applicable	Solid
Water Solubility	Slightly soluble	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
3-Chloroperoxybenzoic acid	1.03	
Benzoic acid, 3-chloro-	2.7	
Vapor Pressure	0.373 hPa @ 25 °C	
Density / Specific Gravity	0.56	
Bulk Density	No data available	
Vapor Density	Not applicable	Solid
Particle characteristics	No data available	

9.2. Other information

Molecular Formula	C7 H5 Cl O3
Molecular Weight	172.57
Evaporation Rate	Not applicable - Solid

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity Yes

10.2. Chemical stability Stable under normal conditions. Organic peroxide. Hazardous decomposition may occur.

10.3. Possibility of hazardous reactions

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Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.
None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat. Temperatures above 35°C - SADT = 55°C.

10.5. Incompatible materials

Reducing Agent. Combustible material. Strong bases.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO₂). Hydrogen chloride gas.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
3-Chloroperoxybenzoic acid	1807 mg/kg (Rat)	-	-
Water	-	-	-

(b) skin corrosion/irritation;

Test method

Category 1 C

Test species

OECD 404

Observational endpoint

rabbit

Causes severe irritation and or burns

(c) serious eye damage/irritation;

Test method

Category 1

Test species

OECD 405

Observation end point

rabbit

CAUSES (SEVERE) EYE BURNS

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Category 1

No information available

(e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

(f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Based on available data, the classification criteria are not met

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(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Not applicable
Solid

Symptoms / effects, both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
3-Chloroperoxybenzoic acid	LC50 (96h): 0.45 mg/l (Cyprinus carpio) OECD Guideline 203		EC50 (72h): 0.49 mg/l (Pseudokirchneriella subcapitata) OECD Guideline 201

Component	Microtox	M-Factor
3-Chloroperoxybenzoic acid		1

12.2. Persistence and degradability

Persistence

Degradation in sewage treatment plant

May persist, based on information available.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely; May have some potential to bioaccumulate

Component	log Pow	Bioconcentration factor (BCF)
3-Chloroperoxybenzoic acid	1.03	No data available
Benzoic acid, 3-chloro-	2.7	No data available

12.4. Mobility in soil

Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.

12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

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12.7. Other adverse effects

Persistent Organic Pollutant
Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number

UN3106

14.2. UN proper shipping name
Technical Shipping Name

ORGANIC PEROXIDE TYPE D, SOLID (3-CHLOROPEROXYBENZOIC ACID)
3-Chloroperoxybenzoic acid

14.3. Transport hazard class(es)

5.2

14.4. Packing group

ADR

14.1. UN number

UN3106

14.2. UN proper shipping name
Technical Shipping Name

ORGANIC PEROXIDE TYPE D, SOLID (3-CHLOROPEROXYBENZOIC ACID)
3-Chloroperoxybenzoic acid

14.3. Transport hazard class(es)

5.2

14.4. Packing group

IATA

14.1. UN number

UN3106

14.2. UN proper shipping name
Technical Shipping Name

ORGANIC PEROXIDE TYPE D, SOLID (3-Chloroperoxybenzoic acid)
3-Chloroperoxybenzoic acid

14.3. Transport hazard class(es)

5.2

14.4. Packing group

14.5. Environmental hazards

Dangerous for the environment
Product is a marine pollutant according to the criteria set by IMDG/IMO

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14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
3-Chloroperoxybenzoic acid	937-14-4	213-322-3	-	-	-	X	KE-05526	X	X
Benzoic acid, 3-chloro-	535-80-8	208-618-4	-	-	X	X	-	X	X
Water	7732-18-5	231-791-2	-	-	X	X	KE-35400	X	-

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
3-Chloroperoxybenzoic acid	937-14-4	X	ACTIVE	X	-	X	X	X
Benzoic acid, 3-chloro-	535-80-8	X	ACTIVE	X	-	X	X	X
Water	7732-18-5	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Not applicable

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
3-Chloroperoxybenzoic acid	937-14-4	-	-	-
Benzoic acid, 3-chloro-	535-80-8	-	-	-
Water	7732-18-5	-	-	-

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
3-Chloroperoxybenzoic acid	937-14-4	Not applicable	Not applicable
Benzoic acid, 3-chloro-	535-80-8	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

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National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = non-hazardous to waters (self classification)

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
3-Chloroperoxybenzoic acid 937-14-4 (70-76)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H242 - Heating may cause a fire
H302 - Harmful if swallowed
H314 - Causes severe skin burns and eye damage
H315 - Causes skin irritation
H317 - May cause an allergic skin reaction
H318 - Causes serious eye damage
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation
H400 - Very toxic to aquatic life
H410 - Very toxic to aquatic life with long lasting effects

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

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ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Creation Date 23-Jan-2009

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Revision Summary Not applicable.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet

Annex to the Safety Data Sheet according to Regulation (EC) No 1907/2006 [REACH]

m-Chloroperoxybenzoic acid (MCPBA) - Exposure Scenarios

CAS No 937-14-4	REACH registration number 01-2120794224-51-0000	EC No 213-322-3
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Exposure Scenarios Overview				
Title	Sector of use	Process category(ies)	Environmental release category	ES Identifier
Formulation of preparations and/or re-packaging	Use at Industrial Sites - Industrial use of the substance as an oxidant in organic synthesis	1, 2, 3, 4, 5, 8a, 8b, 9, 15	ERC6b - Industrial use of reactive processing aids	ES1-F1 MCPBA
Laboratory use	Widespread use by professional workers – Use as a laboratory reagent by professional users	1, 2, 3, 4, 5, 8a, 8b, 9, 15	ERC8b - Wide dispersive indoor use of reactive substances in open systems	ES2-L1 MCPBA

Exposure scenario

ES1 Synthesis/formulating/re-packing of MCPBA - ES1-F1 MCPBA

Section 1 - Identification of the use

Main user group	Industrial uses: Uses of substances as such or in preparations at industrial sites
Type	Worker
Processes, tasks, activities covered	Manufacture or use as an intermediate or process chemical or extraction agent. Loading (including marine vessel/barge, rail/road car and IBC loading) and repacking (including drums and small packs) of substance, including its sampling, storage, unloading distribution and associated laboratory activities
Sector(s) of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Process category(ies)	PROC1 - Use in closed process, no likelihood of exposure PROC2 - Use in closed, continuous process with occasional controlled exposure PROC3 - Use in closed batch process (synthesis or formulation) PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises PROC5 - Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact) PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing) PROC15 - Use as laboratory reagent
Environmental release category(ies)	ERC6b - Industrial use of reactive processing aids

Section 2 - Operational Conditions and Risk Management Measures

Product characteristics

Physical State	Solid paste
pH	4.5
Water Solubility	7961mg/L at 25°C

Vapor Pressure	0.373Pa @ 25 °C
Volatility	Very low
Covers concentrations up to 100 %	
(Imported product is typically 98.8% of 3- Chloroperoxybenzoic acid (72%) balanced with 3-Chlorobenzoic acid (8.8%) and Water (18%))	

Section 2.1 - Control of environmental exposure

Environmental release category(ies)

ERC6b - Industrial use of reactive processing aids

Control of environmental exposure

Not readily biodegradable

Annual amount used in the EU 97.9 t/a

Daily amount per site 3.916 t/d

Environmental factors not influenced by risk management

Emission days	25
Receiving water dilution (fresh or marine)	18000 m³/d

Other operational conditions of use affecting environmental exposure

Emission days	25
Release fraction to air from process (initial release prior to RMM)	$7.3 \times 10^{-30}\%$
Release fraction to wastewater from process (initial release prior to RMM)	94.61%
Release fraction to soil from process (initial release prior to RMM)	5.387%

Conditions and measures related to municipal sewage treatment plant

Assumed domestic sewage treatment plant flow	2000 m³/d
Removal efficiency fraction (offsite; STP)	Standard effectiveness 5.394%

Waste management

Air	ERC release factor 0.1%
Water	$1.75 \times 10^{-50}\%$
	Assumed on-site sewage treatment plant flow
Soil	0.025%

Section 2.2 - Control of worker exposure

General information on exposure estimation

Manufactured and processed at industrial sites in closed continuous processes with either no likelihood of exposure or with only occasional opportunity for exposure in controlled conditions e.g. during maintenance, sampling or discharge of the material.

Control of worker exposure

Process category(ies)	PROC1 - Use in closed process, no likelihood of exposure
Covers concentrations up to	100%
Amounts used	97.9 t/y
Exposure duration	< 8 hour(s)
Use frequency	25 days per year
Indoor/Outdoor use	Indoor
Assumes process temperature up to	$\leq 40^{\circ}\text{C}$
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle substance within a closed system Use of closed transfers of liquids from storage to production equipment (e.g. metered piped or pumped additions) Sample via a closed loop or other system to avoid exposure
Technical conditions and measures to control dispersion from source towards	Undertake operation under enclosed conditions

the worker

Additional good practice advice beyond the REACH Chemical Safety Report Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact Use chemically resistant gloves conforming to EN374 Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices

Process category(ies) PROC2 - Use in closed, continuous process with occasional controlled exposure
 Covers concentrations up to 100%
 Exposure duration < 8 hour(s)
 Indoor/Outdoor use Indoor
 Assumes process temperature up to <=40°C
 Minimum room ventilation rate for handling/application (air changes per hour) 1-3
 Technical conditions and measures to control dispersion from source towards the worker Handle in a fume cupboard or under extract ventilation - efficiency of at least [%]: 90
 Conditions and measures related to personal protection, hygiene and health evaluation Wear chemically resistant gloves (tested to EN374) in combination with specific activity training
 Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
 Additional good practice advice beyond the REACH Chemical Safety Report Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices

Process category(ies) PROC3 - Use in closed batch process (synthesis or formulation)
 Covers concentrations up to 100%
 Exposure duration < 8 hour(s)
 Indoor/Outdoor use Indoor
 Assumes process temperature up to <=40°C
 Minimum room ventilation rate for handling/application (air changes per hour) 1-3
 Organisational measures to prevent /limit releases, dispersion and exposure Handle in a fume cupboard or under extract ventilation
 Local exhaust ventilation - efficiency of at least 90%
 Conditions and measures related to personal protection, hygiene and health evaluation Wear chemically resistant gloves (tested to EN374) in combination with specific activity training
 Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
 Additional good practice advice beyond the REACH Chemical Safety Report Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices

Process category(ies) PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises
 Covers concentrations up to 100%
 Exposure duration < 8 hour(s)
 Indoor/Outdoor use Indoor
 Assumes process temperature up to <=40°C
 Minimum room ventilation rate for handling/application (air changes per hour) 1-3
 Organisational measures to prevent /limit releases, dispersion and exposure Handle in a fume cupboard or under extract ventilation
 Local exhaust ventilation - efficiency of at least 90%
 Conditions and measures related to personal protection, hygiene and health evaluation Wear chemically resistant gloves (tested to EN374) in combination with specific activity training
 Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
 Additional good practice advice beyond the REACH Chemical Safety Report Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices

Process category(ies) PROC5 - Mixing or blending in batch processes for formulation of preparations and articles

Covers concentrations up to	(multistage and/or significant contact) 100%
Exposure duration	Avoid carrying out operation for more than 4 hours
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<= 40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle in a fume cupboard or under extract ventilation Local exhaust ventilation - efficiency of at least 90%
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Assumes a good basic standard of occupational hygiene is implemented

Process category(ies)	PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities
Covers concentrations up to	100%
Exposure duration	< 4 hour(s)
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle in an enclosing hood with exhaust ventilation - efficiency of at least [%]: 90%
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Assumes a good basic standard of occupational hygiene is implemented

Process category(ies)	PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities
Covers concentrations up to	100%
Exposure duration	Avoid carrying out activities involving exposure for more than 4 hours
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Assumes a good basic standard of occupational hygiene is implemented

Process category(ies)	PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing)
Covers concentrations up to	100%
Exposure duration	< 8 hour(s)
Indoor/Outdoor use	Indoor use
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3

Organisational measures to prevent /limit releases, dispersion and exposure	Fill containers/cans at dedicated fill points supplied with local extract ventilation Local exhaust ventilation - efficiency of at least 90%
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Assumes a good basic standard of occupational hygiene is implemented

Process category(ies)	PROC15 - Use as laboratory reagent
Covers concentrations up to	100%
Exposure duration	< 8 hour(s)
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle in a fume cupboard - efficiency of at least [%]: 90
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Assumes a good basic standard of occupational hygiene is implemented

Control of consumer exposure Not intended for consumer use

Section 3 - Exposure estimation

Environment

Environmental release category(ies)

ERC6b - Industrial use of reactive processing aids

Predicted No Effect Concentration (PNEC) - See values below

Fresh water	0.000045 mg/l	Marine water	0.00000045 mg/l
Fresh water sediment	0.000222 mg/kg dw	Marine water sediment	0.0000222 mg/kg dw
Water Intermittent	0.00045 mg/l	Soil (Agriculture)	0.000417 mg/kg dw
Microorganisms in sewage treatment	2 mg/l		

<u>Environment</u>	<u>Predicted exposure level</u>	<u>Risk characterization ratio (RCR)</u>
Freshwater	3.86×10^{-5} mg/l	0.086
Marine water	3.85×10^{-6} mg/l	0.086
Freshwater sediment	1.91×10^{-3} mg/kg dw	0.086
Marine sediment	1.9×10^{-4} mg/kg dw	0.086
Municipal STP	3.24×10^{-4} mg/l	<0.01
Soil	3.14×10^{-4} mg/kg dw	0.075
Air	7.46×10^{-5} mg/m ³	<0.01

Calculation method - EUSES 2.1

Remarks

No significant PEC values are indicated for the regional scale even under the conservative assumptions of the Tier 2 EUSES assessment. All derived PECs are below the relevant PNEC and so no further assessment or refinements are required.

Health

Derived No Effect Level (DNEL) - See table for values

Route of exposure	Acute effects (local)	Acute effects (systemic)	Chronic effects (local)	Chronic effects (systemic)
Oral Dermal Inhalation				0.75 mg/kg bw/day 2.64 mg/m ³

Exposure estimation Workers

Process category(ies)	Exposure route	Predicted exposure level	Risk characterization ratio (RCR)
PROC1 - Use in closed process, no likelihood of exposure	Worker - inhalative, long-term - local and systemic	0.01 mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.04 mg/m ³	
	Worker - dermal, long-term - systemic	0.034 mg/kg bw/day	0.045
	Worker - dermal, long-term - local	9.92 x 10 ⁻³ mg/cm ²	
	Worker - dermal, short-term - local	9.92 x 10 ⁻³ mg/cm ²	
	Worker - combined, long-term - systemic		0.049
PROC2 - Use in closed, continuous process with occasional controlled exposure	Worker - inhalative, long-term - local and systemic	1 x 10 ⁻³ mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	4 x 10 ⁻³ mg/m ³	
	Worker - dermal, long-term - systemic	0.137 mg/kg bw/day	0.183
	Worker - dermal, long-term - local	0.02 mg/cm ²	
	Worker - dermal, short-term - local	0.02 mg/cm ²	
	Worker - combined, long-term - systemic		0.183
PROC3 - Use in closed batch process (synthesis or formulation)	Worker - inhalative, long-term - local and systemic	0.01 mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.04 mg/m ³	
	Worker - dermal, short-term - systemic	0.069 mg/kg bw/day	0.092
	Worker - dermal, long-term - local	0.02 mg/cm ²	
	Worker - dermal, short-term - local	0.02 mg/cm ²	
	Worker - combined, long-term - systemic		0.096
PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises	Worker - inhalative, long-term - local and systemic	0.05 mg/m ³	0.019
	Worker - inhalative, short-term - local and systemic	0.2 mg/m ³	
	Worker - dermal, long-term - systemic	0.343 mg/kg bw/day	0.457
	Worker - dermal, long-term - local	0.05 mg/cm ²	
	Worker - dermal, short-term - local	0.05 mg/cm ²	
	Worker - combined, long-term - systemic		0.476
PROC5 - Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact)	Worker - inhalative, long-term - local and systemic	0.03 mg/m ³	0.011
	Worker - inhalative, short-term - local and systemic	0.2 mg/m ³	
	Worker - dermal, long-term - systemic	0.411 mg/kg bw/day	0.548
	Worker - dermal, long-term - local	0.06 mg/cm ²	
	Worker - dermal, short-term - local	0.06 mg/cm ²	
	Worker - combined, long-term - systemic		0.56
PROC8a - Transfer of substance or	Worker - inhalative, long-term -	0.03 mg/m ³	0.011

preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities	local and systemic		
	Worker - inhalative, short-term - local and systemic	0.2 mg/m ³	
	Worker - dermal, long-term - systemic	0.411 mg/kg bw/day	0.548
	Worker - dermal, long-term - local	0.03 mg/cm ²	
	Worker - dermal, short-term - local	0.03 mg/cm ²	
PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities	Worker - combined, long-term - systemic		0.56
	Worker - inhalative, long-term - local and systemic	3 x 10 ⁻³ mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.02 mg/m ³	
	Worker - dermal, long-term - systemic	0.411 mg/kg bw/day	0.548
	Worker - dermal, long-term - local	0.03 mg/cm ²	
PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing)	Worker - dermal, short-term - local	0.03 mg/cm ²	
	Worker - combined, long-term - systemic		0.549
	Worker - inhalative, short-term - local and systemic	0.01 mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.04 mg/m ³	
	Worker - dermal, long-term - systemic	0.343 mg/kg bw/day	0.457
PROC15 - Use as laboratory reagent	Worker - dermal, long-term - local	0.05 mg/cm ²	
	Worker - dermal, short-term - local	0.05 mg/cm ²	
	Worker - combined, long-term - systemic		0.461
	Worker - inhalative, long-term - local and systemic	0.01 mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.04 mg/m ³	
	Worker - dermal, long-term - systemic	0.068 mg/kg bw/day	0.091
	Worker - dermal, long-term - local	0.02 mg/cm ²	
	Worker - dermal, short-term - local	0.02 mg/cm ²	
	Worker - combined, long-term - systemic		0.094

Calculation method The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated

Exposure route

The hazard assessment is mainly based on the corrosive and sensitising properties of the substance and exposure should be well controlled

Remarks

Predicted exposures are not expected to exceed the DN(M)EL when the risk management measures/operational conditions outlined in section 2 are implemented

Section 4 - Guidance to check compliance with the exposure scenario

Used EUSES model

Used ECETOC TRA model

Further details on scaling and control technologies are provided in SpERC factsheet (<http://cefic.org/en/reach-for-industries-libraries.html>)

Predicted exposures are not expected to exceed the applicable exposure limits (given in section 8 of the SDS) when the

operational conditions/risk management measures given in section 2 are implemented
ECHA guidance for downstream users

Annex to the Safety Data Sheet according to Regulation (EC) No 1907/2006 [REACH]**m-Chloroperoxybenzoic acid (MCPBA) - Exposure Scenarios**

CAS No 937-14-4	REACH registration number 01-2120794224-51-0000	EC No 213-322-3
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Exposure scenario**ES2 Laboratory Use of MCPBA - ES2-L1 MCPBA****Section 1 - Identification of the use**

Main user group	Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Type	Worker
Processes, tasks, activities covered	Laboratory reagent and solvent involving transfer from larger to small containers and vice versa.
Sector(s) of use	SU24 - Scientific research and development SU20 - Health services SU0 - Other
Process category(ies)	PROC1 - Use in closed process, no likelihood of exposure PROC2 - Use in closed, continuous process with occasional controlled exposure PROC3 - Use in closed batch process (synthesis or formulation) PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises PROC5 - Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact) PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing) PROC15 - Use as laboratory reagent
Environmental release category(ies)	ERC8b - Wide dispersive indoor use of reactive substances in open systems

Section 2 - Operational Conditions and Risk Management Measures**Product characteristics**

Physical State	Solid paste
pH	4.5
Water Solubility	7961mg/L at 25°C
Vapor Pressure	0.373Pa @ 25 °C
Volatility	Very low

Covers concentrations up to 100 %

(Imported product is typically 98.8% of 3- Chloroperoxybenzoic acid (72%) balanced with 3-Chlorobenzoic acid (8.8%) and Water (18%))

Section 2.1 - Control of environmental exposure**Environmental release category(ies)**

ERC8b - Wide dispersive indoor use of reactive substances in open systems

Control of environmental exposure

Not readily biodegradable

Annual amount used in the EU 97.9 t/a

Daily amount per site 5.38×10^{-5} t/d

Other operational conditions of use affecting environmental exposure

Release fraction to air from process (initial release prior to RMM) 0.1%

Release fraction to wastewater from process (initial release prior to RMM) 2%

Release fraction to soil from process (initial release prior to RMM) 0%

Conditions and measures related to municipal sewage treatment plant

Assumed domestic sewage treatment plant flow 2000 m³/d

Removal efficiency fraction (offsite; STP) Standard effectiveness 5.394%

Section 2.2 - Control of worker exposure

General information on exposure estimation

Local exhaust ventilation (LEV) usually required for indoor industrial use.

Control of worker exposure

Process category(ies)	PROC1 - Use in closed process, no likelihood of exposure
Covers concentrations up to	100%
Exposure duration	< 8 hour(s)
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle substance within a closed system Sample via a closed loop or other system to avoid exposure
Technical conditions and measures to control dispersion from source towards the worker	Undertake operation under enclosed conditions
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices

Process category(ies)	PROC2 - Use in closed, continuous process with occasional controlled exposure
Covers concentrations up to	100%
Exposure duration	< 8 hour(s)
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Technical conditions and measures to control dispersion from source towards the worker	Handle in a fume cupboard or under extract ventilation - efficiency of at least [%]: 80
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there

Additional good practice advice beyond the REACH Chemical Safety Report	is potential for direct contact Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices -----
Process category(ies)	PROC3 - Use in closed batch process (synthesis or formulation)
Covers concentrations up to	100%
Exposure duration	< 8 hour(s)
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle in a fume cupboard or under extract ventilation Local exhaust ventilation - efficiency of at least 80%
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices -----
Process category(ies)	PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises
Covers concentrations up to	100%
Exposure duration	< 4 hour(s)
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle in a fume cupboard or under extract ventilation Local exhaust ventilation - efficiency of at least 80%
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices -----
Process category(ies)	PROC5 - Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact)
Covers concentrations up to	100%
Exposure duration	Avoid carrying out operation for more than 1 hour
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<= 40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle in a fume cupboard or under extract ventilation Local exhaust ventilation - efficiency of at least 80%
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Assumes a good basic standard of occupational hygiene is implemented -----
Process category(ies)	PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities

Covers concentrations up to	100%
Exposure duration	< 1 hour(s)
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle in an enclosing hood with exhaust ventilation - efficiency of at least [%]: 80%
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Assumes a good basic standard of occupational hygiene is implemented

Process category(ies)	PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities
Covers concentrations up to	100%
Exposure duration	Avoid carrying out operation for more than 1 hour
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Handle in an enclosing hood with exhaust ventilation - efficiency of at least [%]: 80
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Assumes a good basic standard of occupational hygiene is implemented

Process category(ies)	PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing)
Covers concentrations up to	100%
Exposure duration	< 4 hour(s)
Indoor/Outdoor use	Indoor use
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per hour)	1-3
Organisational measures to prevent /limit releases, dispersion and exposure	Fill containers/cans at dedicated fill points supplied with local extract ventilation Local exhaust ventilation - efficiency of at least 80%
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact
Additional good practice advice beyond the REACH Chemical Safety Report	Assumes a good basic standard of occupational hygiene is implemented

Process category(ies)	PROC15 - Use as laboratory reagent
Covers concentrations up to	100%
Exposure duration	< 8 hour(s)
Indoor/Outdoor use	Indoor
Assumes process temperature up to	<=40°C
Minimum room ventilation rate for handling/application (air changes per	1-3

hour)

Organisational measures to prevent /limit releases, dispersion and exposure

Handle in a fume cupboard - efficiency of at least [%]: 90

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training

Use chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact

Additional good practice advice beyond the REACH Chemical Safety Report Assumes a good basic standard of occupational hygiene is implemented

Control of consumer exposure

Not intended for consumer use

Section 3 - Exposure estimation

Environment

Environmental release category(ies)

ERC8b - Wide dispersive indoor use of reactive substances in open systems

Predicted No Effect Concentration (PNEC) - See values below

Fresh water	0.000045 mg/l	Marine water	0.00000045 mg/l
Fresh water sediment	0.000222 mg/kg dw	Marine water sediment	0.0000222 mg/kg dw
Water Intermittent	0.00045 mg/l	Soil (Agriculture)	0.000417 mg/kg dw
Microorganisms in sewage treatment	2 mg/l		

Environment	Predicted exposure level	Risk characterization ratio (RCR)
Freshwater	5.72×10^{-5} mg/l	0.127
Marine water	5.71×10^{-6} mg/l	0.127
Freshwater sediment	2.82×10^{-3} mg/kg dw	0.127
Marine sediment	2.81×10^{-4} mg/kg dw	0.127
Municipal STP	5.09×10^{-4} mg/l	<0.01
Soil	2.08×10^{-4} mg/kg dw	0.05

Calculation method - EUSES 2.1

Remarks

No significant PEC values are indicated for the regional scale even under the conservative assumptions of the Tier 2 EUSES assessment. All derived PECs are below the relevant PNEC and so no further assessment or refinements are required.

Health

Derived No Effect Level (DNEL) - See table for values

Route of exposure	Acute effects (local)	Acute effects (systemic)	Chronic effects (local)	Chronic effects (systemic)
Oral				
Dermal				0.75 mg/kg bw/day
Inhalation				2.64 mg/m ³

Exposure estimation

Workers

Process category(ies)	Exposure route	Predicted exposure level	Risk characterization ratio (RCR)
PROC1 - Use in closed process, no likelihood of exposure	Worker - inhalative, long-term - local and systemic	0.01 mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.04 mg/m ³	
	Worker - dermal, long-term - systemic	0.034 mg/kg bw/day	0.045
	Worker - dermal, long-term - local	9.92×10^{-3} mg/cm ²	
	Worker - dermal, short-term - local	9.92×10^{-3} mg/cm ²	

	Worker - combined, long-term - systemic		0.049
PROC2 - Use in closed, continuous process with occasional controlled exposure	Worker - inhalative, long-term - local and systemic	$2 \times 10^{-3} \text{ mg/m}^3$	<0.01
	Worker - inhalative, short-term - local and systemic	$8 \times 10^{-3} \text{ mg/m}^3$	
	Worker - dermal, long-term - systemic	0.274 mg/kg bw/day	0.183
	Worker - dermal, long-term - local	0.04 mg/cm ²	
	Worker - dermal, short-term - local	0.04 mg/cm ²	
	Worker - combined, long-term - systemic		0.366
PROC3 - Use in closed batch process (synthesis or formulation)	Worker - inhalative, long-term - local and systemic	0.02 mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.08 mg/m ³	
	Worker - dermal, short-term - systemic	0.138 mg/kg bw/day	0.184
	Worker - dermal, long-term - local	0.04 mg/cm ²	
	Worker - dermal, short-term - local	0.04 mg/cm ²	
	Worker - combined, long-term - systemic		0.192
PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises	Worker - inhalative, long-term - local and systemic	0.12 mg/m ³	0.045
	Worker - inhalative, short-term - local and systemic	0.8 mg/m ³	
	Worker - dermal, long-term - systemic	0.412 mg/kg bw/day	0.549
	Worker - dermal, long-term - local	0.06 mg/cm ²	
	Worker - dermal, short-term - local	0.06 mg/cm ²	
	Worker - combined, long-term - systemic		0.594
PROC5 - Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact)	Worker - inhalative, long-term - local and systemic	0.04 mg/m ³	0.015
	Worker - inhalative, short-term - local and systemic	0.8 mg/m ³	
	Worker - dermal, long-term - systemic	0.274 mg/kg bw/day	0.366
	Worker - dermal, long-term - local	0.04 mg/cm ²	
	Worker - dermal, short-term - local	0.04 mg/cm ²	
	Worker - combined, long-term - systemic		0.381
PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities	Worker - inhalative, long-term - local and systemic	0.02 mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.4 mg/m ³	
	Worker - dermal, long-term - systemic	0.274 mg/kg bw/day	0.366
	Worker - dermal, long-term - local	0.02 mg/cm ²	
	Worker - dermal, short-term - local	0.02 mg/cm ²	
	Worker - combined, long-term - systemic		0.373
PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities	Worker - inhalative, long-term - local and systemic	0.02 mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.4 mg/m ³	
	Worker - dermal, long-term - systemic	0.274 mg/kg bw/day	0.366

	Worker - dermal, long-term - local	0.02 mg/cm ²	
	Worker - dermal, short-term - local	0.02 mg/cm ²	
	Worker - combined, long-term - systemic		0.373
PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing)	Worker - inhalative, short-term - local and systemic	0.06 mg/m ³	0.023
	Worker - inhalative, short-term - local and systemic	0.4 mg/m ³	
	Worker - dermal, long-term - systemic	0.412 mg/kg bw/day	0.549
	Worker - dermal, long-term - local	0.06 mg/cm ²	
	Worker - dermal, short-term - local	0.06 mg/cm ²	
	Worker - combined, long-term - systemic		0.572
PROC15 - Use as laboratory reagent	Worker - inhalative, long-term - local and systemic	0.02 mg/m ³	<0.01
	Worker - inhalative, short-term - local and systemic	0.08 mg/m ³	
	Worker - dermal, long-term - systemic	0.068 mg/kg bw/day	0.091
	Worker - dermal, long-term - local	0.02 mg/cm ²	
	Worker - dermal, short-term - local	0.02 mg/cm ²	
	Worker - combined, long-term - systemic		0.098

Calculation method

The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated

Exposure route

The hazard assessment is mainly based on the corrosive and sensitising properties of the substance and exposure should be well controlled

Remarks

Predicted exposures are not expected to exceed the DN(M)EL when the risk management measures/operational conditions outlined in section 2 are implemented

Section 4 - Guidance to check compliance with the exposure scenario

Used EUSES model

Used ECETOC TRA model

Further details on scaling and control technologies are provided in SpERC factsheet (<http://cefic.org/en/reach-for-industries-libraries.html>)

Predicted exposures are not expected to exceed the applicable exposure limits (given in section 8 of the SDS) when the operational conditions/risk management measures given in section 2 are implemented

ECHA guidance for downstream users